



# IndianOil

## JOURNEY RISK MANAGEMENT (JRM) STUDY

### Salem Terminal to ADHOC SREE PALANIVEL TRADERS UNIT II

#### Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

#### Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

#### Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

## **Compliance with the Road Transport Safety Policy (RTSP)**

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

## **Emergency Preparedness and Response**

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

## **Environmental Considerations**

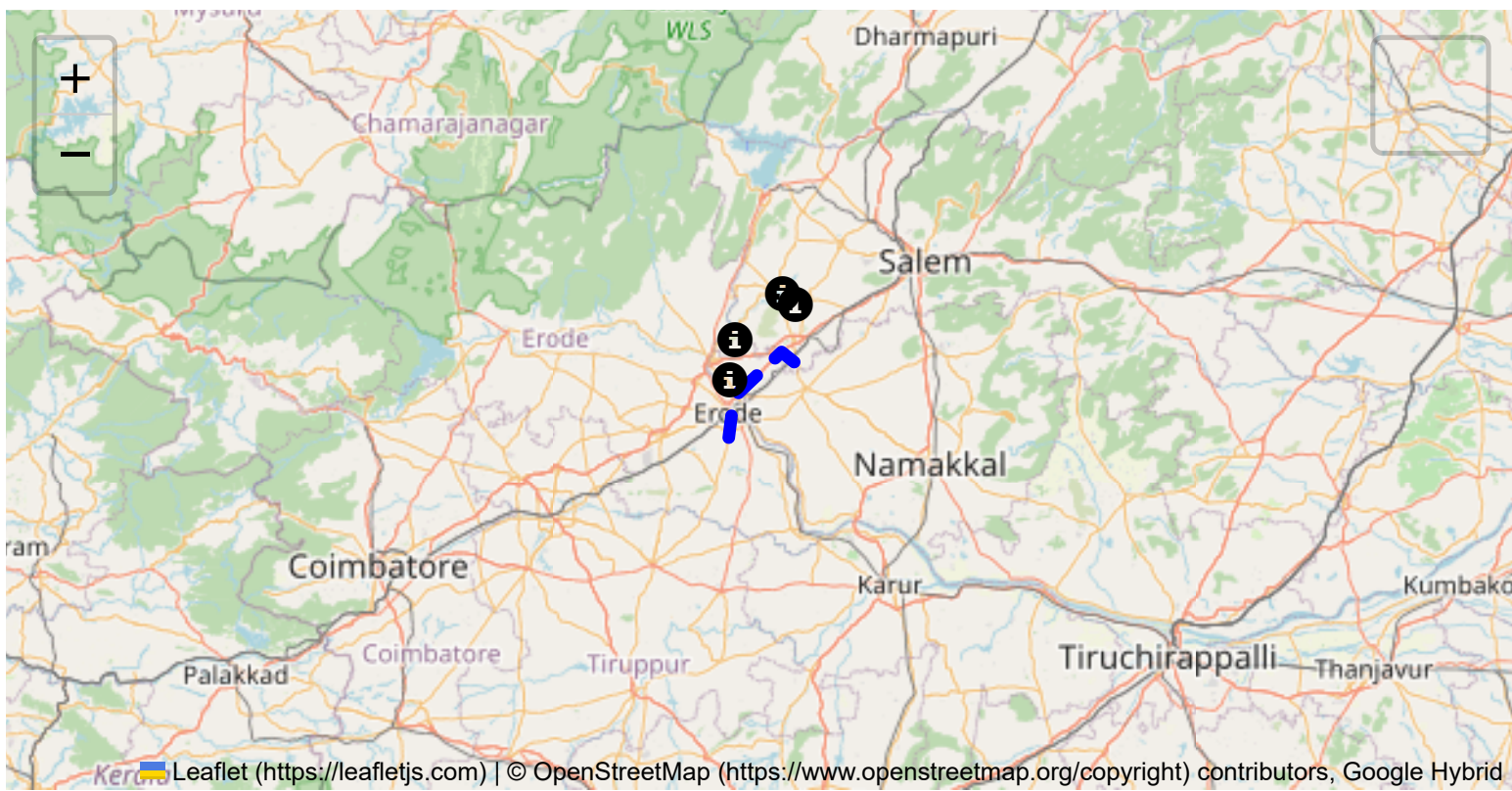
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

## **Journey Risk Mitigation**

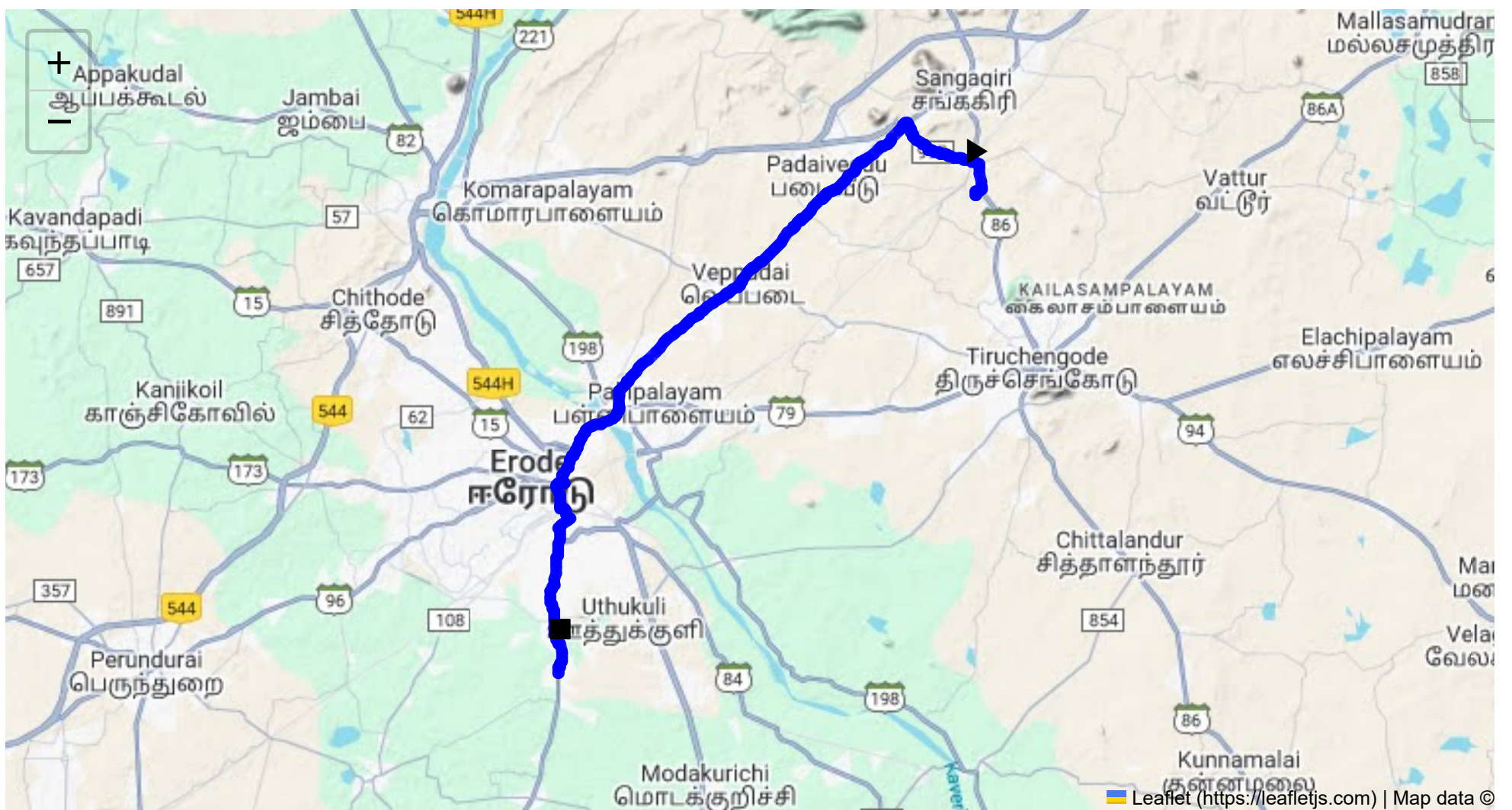
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

## **Compliance with Government Directives**

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



**Route Summary:**  
**Total Distance: 32.56 km**  
**Estimated Duration: 0.9 hours**  
**Adjusted Duration (Heavy Vehicle): 1.1 hours**  
**Start: (11.4381, 77.8734)**  
**End: (11.273925, 77.726665)**



**Welcome to the Journey Risk Management Study**

## Route Safety Analysis

### 1. Overview of the Route Map



The route from Sangagiri to Kaspapettai via the mentioned points covers approximately 32.56 kilometers and takes roughly 0.90 hours for heavy vehicles carrying hazardous materials. Starting from CVQF+23W in Sangagiri, the path leads to FR4V+R5 Alex in Thomas Colony, passing through Erode along the Attur - Rasipuram - Erode Road (SH 83A) before reaching 7PFG+CQR in Kaspapettai.

## 2. Typical Weather Conditions and Hazards

The region experiences a tropical climate, with temperatures ranging from moderate to hot. The monsoon season (June to September) may lead to heavy rains, potentially causing floods and slippery road conditions. During the summer months (March to June), high temperatures can cause heat stress and vehicle overheating.

## 3. Traffic Patterns and Congestion-Prone Areas

Traffic tends to peak during morning (7:30 - 9:30 AM) and evening (5:30 - 8:00 PM) hours, especially near urban centers like Erode. Congestion is common in Erode, particularly on major roads entering and exiting the city. The junctions along SH 83A can also experience heavy traffic.

## 4. Road Quality and Infrastructure

Generally, SH 83A is well-maintained, but there are patches where potholes and uneven surfaces could be problematic. These issues are more evident in rural areas between major towns. Continuous maintenance is required to avoid potential hazards due to poor road quality.

## 5. Alternative Routes for Emergencies

In case of emergencies, an alternative route via rural roads parallel to SH 83A may be used, though they generally are less maintained. Another alternative is taking NH79 which bypasses Erode to the north, although it would add distance to the journey.

## 6. Local Regulations on Hazardous Material Transport

Transportation of hazardous materials is regulated by national and local guidelines. Drivers are required to follow Indian Motor Vehicles Act provisions, ensure proper labeling, and maintain necessary documentation. Night travel for vehicles carrying hazardous materials may be restricted in some urban zones.

## 7. Historical Incidents Involving Heavy Vehicles

Historically, incidents in the area often occur due to vehicle breakdowns or accidents near congestion points in Erode. Major accidents involving hazardous materials are less frequent but necessitate stringent safety protocols.

## 8. Environmental Considerations and Sensitive Areas

The route passes through both urban and rural areas. Sensitive agricultural zones along the route require careful navigation to prevent environmental contamination. Local wildlife crossings near forested areas

should be noted for animal safety.

## 9. Communication Coverage

Communication coverage is generally good along SH 83A, but some rural segments may have dead zones. Drivers should ensure communication devices are fully charged and consider additional equipment like satellite phones for emergencies.

## 10. Estimated Emergency Response Times

Emergency response within cities like Erode is relatively quick, often within 15-30 minutes. Rural segments may experience delayed response times of up to 45-60 minutes. It is vital to know the locations of nearby hospitals and fire stations.

## 11. Overall Summary of Risk Assessment

While the primary route is generally safe, there are potential risks associated with traffic congestion, weather conditions, road integrity, and communication coverage. Preparation includes planning for alternative routes, awareness of local regulations, and effective communication measures. Ensuring vehicle maintenance, adhering to speed limits, and staying informed about weather conditions will mitigate many risks associated with this route.

### Risk Assessment - Turns

|    | Risk Type  | Risk Level | Coordinates        | Speed Limit |
|----|------------|------------|--------------------|-------------|
| 0  | Roundabout | High       | 11.37086, 77.74828 | 15 KM/Hr    |
| 1  | Turn       | High       | 11.43811, 77.87348 | 15 KM/Hr    |
| 2  | Turn       | Medium     | 11.43956, 77.87340 | 30 KM/Hr    |
| 3  | Turn       | High       | 11.43968, 77.87345 | 15 KM/Hr    |
| 4  | Turn       | High       | 11.44029, 77.87544 | 15 KM/Hr    |
| 5  | Turn       | Medium     | 11.44898, 77.87410 | 30 KM/Hr    |
| 6  | Turn       | Medium     | 11.45357, 77.85789 | 30 KM/Hr    |
| 7  | Turn       | Medium     | 11.45352, 77.85698 | 30 KM/Hr    |
| 8  | Turn       | Medium     | 11.46303, 77.84945 | 30 KM/Hr    |
| 9  | Turn       | Medium     | 11.46318, 77.84913 | 30 KM/Hr    |
| 10 | Turn       | Medium     | 11.46312, 77.84886 | 30 KM/Hr    |
| 11 | Turn       | Medium     | 11.36626, 77.74788 | 30 KM/Hr    |
| 12 | Blind Spot | Blind Spot | 11.33920, 77.72902 | 10 KM/Hr    |

|    | Risk Type | Risk Level | Coordinates        | Speed Limit |
|----|-----------|------------|--------------------|-------------|
| 13 | Turn      | High       | 11.33914, 77.72659 | 15 KM/Hr    |
| 14 | Turn      | High       | 11.33890, 77.72637 | 15 KM/Hr    |
| 15 | Turn      | Medium     | 11.33093, 77.72776 | 30 KM/Hr    |
| 16 | Turn      | Medium     | 11.33076, 77.72884 | 30 KM/Hr    |
| 17 | Turn      | High       | 11.32694, 77.73073 | 15 KM/Hr    |
| 18 | Turn      | Medium     | 11.32427, 77.72730 | 30 KM/Hr    |
| 19 | Turn      | Medium     | 11.32401, 77.72723 | 30 KM/Hr    |

## Emergency Locations

|    | type     | name                                | coordinates               | speed_limit | risk_level |
|----|----------|-------------------------------------|---------------------------|-------------|------------|
| 1  | hospital | Cheran Hospital                     | 11.3680302,<br>77.7481976 | 30 km/h     | Medium     |
| 2  | hospital | Sri Kumaran E N T Hospital          | 11.3657814,<br>77.7477763 | 30 km/h     | Medium     |
| 3  | pharmacy | Pharmacy                            | 11.3652161,<br>77.7486375 | 30 km/h     | Medium     |
| 4  | hospital | Visaalam Hospitals                  | 11.3651528,<br>77.747808  | 30 km/h     | Medium     |
| 5  | hospital | Sri Krishna Hospital                | 11.3656648,<br>77.7477783 | 30 km/h     | Medium     |
| 6  | police   | Police Station                      | 11.3630177,<br>77.7476505 | 30 km/h     | Medium     |
| 7  | hospital | Government Hospital                 | 11.3608508,<br>77.747453  | 30 km/h     | Medium     |
| 9  | hospital | Sri Hari Sakthivel Hospital         | 11.3526901,<br>77.7323055 | 30 km/h     | Medium     |
| 10 | hospital | Senthil Prakash Hospital            | 11.352639, 77.732267      | 30 km/h     | Medium     |
| 11 | hospital | Dhanvantri Critical Care Center     | 11.3504266,<br>77.7311274 | 30 km/h     | Medium     |
| 12 | hospital | V K Hospital                        | 11.3513792,<br>77.7313976 | 30 km/h     | Medium     |
| 13 | hospital | Indrajith Yoga Nature Cure Hospital | 11.3499973,<br>77.732974  | 30 km/h     | Medium     |
| 15 | hospital | Dhanvantri Critical Care Centre     | 11.35099, 77.730508       | 30 km/h     | Medium     |

|    | type         | name                                       | coordinates               | speed_limit | risk_level |
|----|--------------|--------------------------------------------|---------------------------|-------------|------------|
| 16 | hospital     | Kumudha Hospital                           | 11.3467172,<br>77.7306797 | 30 km/h     | Medium     |
| 17 | hospital     | City Hospital                              | 11.3451151,<br>77.7300887 | 30 km/h     | Medium     |
| 18 | clinic       | K.B.N. Nursing Home                        | 11.344835,<br>77.7310771  | 30 km/h     | Medium     |
| 19 | clinic       | Kpn Nursing Home                           | 11.3437022,<br>77.7304525 | 30 km/h     | Medium     |
| 20 | hospital     | Kaviraj Ayurveda Hospital                  | 11.3400349,<br>77.7298199 | 30 km/h     | Medium     |
| 21 | hospital     | Dr Kokilasekar Nursing Hospital            | 11.3395874,<br>77.7269838 | 30 km/h     | Medium     |
| 22 | clinic       | M.R Nursing Home                           | 11.3386746,<br>77.7294262 | 30 km/h     | Medium     |
| 24 | hospital     | Government Maternity Hospital,<br>Agathiar | 11.33716, 77.727075       | 30 km/h     | Medium     |
| 25 | clinic       | Naseer Nursing Home                        | 11.3364485,<br>77.7271691 | 30 km/h     | Medium     |
| 26 | hospital     | Vikram Hospital, Erode                     | 11.3356411,<br>77.7280569 | 30 km/h     | Medium     |
| 27 | hospital     | Shabeer Hospital                           | 11.33529, 77.7285236      | 30 km/h     | Medium     |
| 28 | hospital     | Dr. R.M. Habibullah Hospital               | 11.335246,<br>77.7266028  | 30 km/h     | Medium     |
| 29 | hospital     | Kumarasamy Hospital                        | 11.334273,<br>77.7259118  | 30 km/h     | Medium     |
| 30 | hospital     | Thangam Hospital, Erode                    | 11.3340961,<br>77.7262437 | 30 km/h     | Medium     |
| 31 | hospital     | J.B. Clinic And Hospital                   | 11.3351769,<br>77.7282031 | 30 km/h     | Medium     |
| 32 | hospital     | Ck Hospital                                | 11.3326562,<br>77.7282011 | 30 km/h     | Medium     |
| 33 | hospital     | Immick Hospital                            | 11.332617, 77.728073      | 30 km/h     | Medium     |
| 34 | fire_station | Erode District Fire Station                | 11.3326366,<br>77.7272532 | 30 km/h     | Medium     |
| 35 | hospital     | Jai Maaruthi Hospital                      | 11.332, 77.727272         | 30 km/h     | Medium     |
| 36 | hospital     | Shubiksha Hospital                         | 11.3318174,<br>77.7299544 | 30 km/h     | Medium     |
| 37 | hospital     | JJ hospital                                | 11.33168, 77.728823       | 30 km/h     | Medium     |

|    | type     | name                      | coordinates               | speed_limit | risk_level |
|----|----------|---------------------------|---------------------------|-------------|------------|
| 38 | hospital | Bethesda Hospital         | 11.3314183,<br>77.7266231 | 30 km/h     | Medium     |
| 39 | hospital | Lotus Hospital            | 11.3266236,<br>77.7306301 | 30 km/h     | Medium     |
| 40 | hospital | Lotus Hospital            | 11.3259471,<br>77.7302149 | 30 km/h     | Medium     |
| 41 | clinic   | Monika Diabetes Centre    | 11.3229084,<br>77.7269179 | 30 km/h     | Medium     |
| 42 | hospital | Sri Jothi Hospitals       | 11.3217208,<br>77.7275956 | 30 km/h     | Medium     |
| 43 | clinic   | Vasanth Clinic            | 11.3195299,<br>77.727194  | 30 km/h     | Medium     |
| 44 | clinic   | Pulze Health Clinic       | 11.3179874,<br>77.7288603 | 30 km/h     | Medium     |
| 45 | hospital | Agathiyar Siddha Hospital | 11.3174229,<br>77.7254599 | 30 km/h     | Medium     |
| 46 | hospital | Om Shakthi Hospital       | 11.3108011,<br>77.7259973 | 30 km/h     | Medium     |

### Crowded Spots

|    | type        | name                                       | coordinates               | speed_limit | risk_level |
|----|-------------|--------------------------------------------|---------------------------|-------------|------------|
| 0  | school      | Ssri Valliappa Vidhayalayam mat hr school  | 11.4004878,<br>77.7794381 | 30 km/h     | Medium     |
| 8  | marketplace | Weekly Market                              | 11.3605669,<br>77.7488199 | 30 km/h     | Medium     |
| 14 | school      | Municipality Girls Higher secondary school | 11.3495618,<br>77.7325011 | 30 km/h     | Medium     |
| 23 | school      | E.V.R. Govt Girls Hr. Sec School           | 11.3371265,<br>77.7266283 | 30 km/h     | Medium     |

### Route Photos of Risky Spots





**Risk Type:** Roundabout

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.37086, 77.74828



**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.44029, 77.87544





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.44898, 77.87410



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.45357, 77.85789



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.45352, 77.85698





**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.46303, 77.84945



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.46318, 77.84913





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.46312, 77.84886



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.36626, 77.74788



**Risk Type:** Blind Spot  
**Risk Level:** Blind Spot  
**Speed Limit:** 10 KM/Hr  
**Coordinates:** 11.33920, 77.72902





Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.33914, 77.72659



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.33890, 77.72637





**Risk Type: Turn**

**Risk Level: Medium**

**Speed Limit: 30 KM/Hr**

**Coordinates: 11.33093, 77.72776**



**Risk Type: Turn**

**Risk Level: Medium**

**Speed Limit: 30 KM/Hr**

**Coordinates: 11.33076, 77.72884**



**Risk Type: Turn**

**Risk Level: High**

**Speed Limit: 15 KM/Hr**

**Coordinates: 11.32694, 77.73073**





**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.32427, 77.72730



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.32401, 77.72723

